

# Muhammad Rehan Khalid

AI Engineer specializing in LLM systems, Agentic AI, and scalable production ML platforms

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I have a valid work permit for Germany. My notice period is 30 days.



## TECHNICAL SKILLS

AI Engineer (5+ years of project experience) **Programming Languages:** Python **AI / Machine Learning:** Pytorch | Machine Learning | Deep Learning | NLP | LLMs | Huggingface | scikit-learn | RAG | Transformers | Prompt Engineering **LLM & Agent Frameworks:** LangGraph | LlamaIndex | LangChain | OpenAI API | DeepEval | RAGAS | Hugging Face Transformers **Backend & APIs:** FastAPI | REST APIs **Data & Databases:** Redis | Pyspark | PostgreSQL | Vector Databases | Weaviate **Cloud Platforms:** AWS | Azure **DevOps & MLOps:** Docker | Kubernetes | CI/CD | MLflow | Kubeflow **Streaming & Messaging:** Kafka **Version Control & Collaboration:** Git | Agile

## WORK EXPERIENCE

07/2025 - today

### Data Science Intern - LLM

CHECK24 Vergleichsportal GmbH, München, Germany

**Project:** DocGPT Document RAG Platform – AI based document extraction and chat system used by CHECK24 verticals

- Implemented a Retrieval-Augmented Generation layer for the DocGPT extraction and chat platform, enabling long-document support beyond LLM context limits and onboarding additional CHECK24 verticals.
- Built multi-format ingestion and indexing for PDF, DOCX, PPT, web and Notion sources with cleaning, deduplication, metadata tagging and chunking; stored embeddings in Weaviate for fast semantic retrieval.
- Developed hybrid retrieval workflows with BM25/vector search, re-ranking and query rewriting, improving grounded Q&A and structured invoice/contract extraction into normalized JSON.
- Built an agentic scan-preprocessing pipeline for CHECK24 Autoteile (Reifen), improving OCR/LLM input quality with dewarping, deblurring and lighting/shadow correction.
- Developed a real-time monitoring dashboard for request volume, latency and error rates, reducing 5xx errors by about 15% and helping scale DocGPT extraction usage to 80K+ weekly API requests.

**Technologies used:** Python | LLMs | RAG | Weaviate | BM25 | Vector Search | Re-ranking | OCR | Monitoring

12/2024 - 06/2025

### Research Assistant - LLM

Fraunhofer IAIS, Sankt Augustin, Germany

**Project:** German Legal AI Assistant - Germany-based legal QA platform built on German legal documents

- Generated a high-quality instruction dataset of 7,000+ QA pairs from German legal documents using LLM-assisted curation.
- Fine-tuned open-source LLMs with PEFT (LoRA, QLoRA) using Unsloth and PyTorch, improving response relevance by 20%.
- Built and optimized RAG pipelines, improving context recall by 16% and precision by 10%; fine-tuned cross-encoder re-rankers to improve search relevance by 15%.
- Automated evaluation frameworks with RAGAS, DeepEval and LLM-as-a-Judge for repeatable benchmarking.

**Technologies used:** Python | LLMs | PEFT | LoRA | QLoRA | Unsloth | PyTorch | RAGAS | DeepEval | RAG

01/2024 - 11/2024

### Research Assistant - MLOps

Fraunhofer IVI, Dresden, Germany

**Project:** Autonomous Driving Signal Monitoring Platform - Germany-based geospatial MLOps system for AD-drop and RSU status monitoring

- Built Grafana dashboards for Autonomous Driving signal quality and AD-drop monitoring using OpenStreetMap data, enabling real-time outage alerts.

- Enriched TimescaleDB datasets with Roadside Unit and bus-stop metadata, reducing query latency by 40%.
  - Managed and optimized TimescaleDB data workflows, implementing retention policies and integrity constraints to improve reliability.
- Technologies used:** Python | Grafana | TimescaleDB | PostgreSQL | OpenStreetMap | Geospatial Data | Monitoring

12/2021 - 08/2023

**Data Analyst**  
i2c Inc., Lahore, Pakistan  
**Project:** AI Fraud Detection Platform - USA-based fraud detection engine for real-time transaction monitoring used by financial institutions

- Led fraud detection projects using ensemble models (XGBoost, LightGBM) and neural networks (TabNet, RNNs), capturing 40% more fraud at 1:5 false-positive odds.
- Implemented real-time fraud prevention models, reducing annual losses by \$150K and improving detection response for high-risk transactions.
- Engineered profiling pipelines generating rolling-window features from minutes to 30 days and reduced class imbalance from 1/1000 to 1/100 through data sampling.
- Enhanced Account Takeover fraud detection by integrating behavioral signals, reducing ATO cases by 15%.

**Technologies used:** Python | XGBoost | LightGBM | TabNet | RNNs | Fraud Detection | Feature Engineering

08/2020 - 12/2021

**Data Scientist**  
Afiniti, Lahore, Pakistan  
**Project:** Customer Propensity Modeling Platform - customer ranking models for revenue optimization

- Developed LightGBM-based customer propensity models for customer ranking, achieving a 10% revenue increase.
- Conducted A/B testing and ROI analyses, contributing to a 6% uplift in client-side sales.

**Technologies used:** Python | LightGBM | Machine Learning | A/B Testing | ROI Analysis

EDUCATION

10/2024 - 04/2026

MSc Applied Data Science  
George August Universität, Göttingen, German

10/2016 - 08/2020

BS Math and Economics  
LUMS, Lahore, Pakistan

LANGUAGE SKILLS

English (Mother tongue) | German (B1, currently learning)

HOBBIES

Squash | Reading | Baking